The Most Promising BiCRS Pathways in the United Kingdom

Biomass Carbon Removal and Storage (BiCRS) converts waste and residual biomass into durable CO_2 removal — either by capturing and geologically storing the biogenic CO_2 released when biomass is used for energy or fuel, or by locking the carbon away directly. This brief summarises where the UK's most promising opportunities lie, drawn from the interactive **UK BiCRS Atlas** (https://hausfath.github.io/bicrs-map-uk/), which resolves feedstock, storage, and best-use recommendations to each of the UK's 41 NUTS-2 / ITL-2 regions.

The Atlas ranks pathways using Frontier's procurement KPIs — **carbon-removal efficiency first, then emissions-avoiding co-products, then wider co-benefits** — screened against feedstock moisture and density, distance and cost to reach CO_2 storage, soil-nutrient status, and whether a facility suitable for retrofit already exists nearby. Purpose-grown energy crops are excluded by design; the focus is genuinely residual and waste biomass.

Headline: how much durable removal the UK's residual biomass supports depends heavily on **what kind of CO_2 storage is available**, so we frame two scenarios:

- **Scenario A — gaseous-CO_2 storage only (~8.1 Mt CO_2/yr).** The UK's developed and in-construction stores (the offshore CCS hubs) accept **gaseous / dense-phase CO_2 only**. This scenario counts just the pathways that capture CO_2 (BECCS, waste-to-energy + CCS, anaerobic digestion + CCS) plus storage-independent biochar. It is the near-term, shovel-ready floor.
- **Scenario B — with salt-cavern injection storage (~14.9 Mt CO_2/yr).** If the UK develops **salt caverns** to store biomass slurry and bio-oil (a recognised method, but no UK site is yet permitted for it), the wet livestock regions and dry-residue regions far from a capture anchor gain a pathway, lifting potential by **~6.8 Mt CO_2/yr (+84%)**.

Both are screening estimates of what a mature, right-sized UK BiCRS sector could deliver from wastes and residues alone, before any dedicated energy crops. The UK is unusually well positioned on the capture side: it pairs a modest but real residue base with the largest offshore CO_2 storage resource in Europe and two storage hubs already in construction.

1. Feedstock availability

The UK is not a biomass-rich country by land area, but its **waste streams are substantial and highly concentrated near population and industry** — favourable for BiCRS, which rewards feedstock that is already aggregated. Recoverable UK totals (Mt/yr, oven-dry except MSW):

Feedstock	Recoverable supply	Character & best fit
Municipal solid waste (total)	~25.7 Mt/yr (~55% biogenic)	Concentrated in cities; already collected → **WtE + CCS**
Animal manure	~12.0 Mt/yr	Wet, distributed across the livestock west/north → **AD + CCS** or **HTL bio-oil**
Agricultural residues	~6.3 Mt/yr	Straw/stover, concentrated in the arable east → **BECCS / injection**
Human / WWTP biosolids	~1.0 Mt/yr	Wet, at large sewage works → co-digestion / injection
Forestry residues	~0.5 Mt/yr	Small; the UK is import-dependent for wood

Two features stand out. First, **MSW and manure dominate the tonnage** — both wet or heterogeneous streams that suit anaerobic digestion, waste-to-energy, or hydrothermal routes rather than dry combustion. Second, the geography is clean: **arable dry residues cluster in East Anglia and the east**, while **wet manure dominates the livestock west and north** (Wales, South West England, Scotland, Northern Ireland). That split largely determines the recommended pathway region-by-region.

Manure and crop-residue supply are distributed using the **UK June Agricultural Census** (housed cattle/pig/poultry numbers for manure; arable-land area for straw), so recoverable manure tracks where livestock actually are — near-zero in London and the conurbations, concentrated in Northern Ireland, West Wales, the South West, North Yorkshire and southern Scotland.

2. Retrofit opportunities

The UK's strongest near-term advantage is a **large existing biomass and waste-combustion fleet that can be retrofitted with capture**, avoiding greenfield build. The Atlas maps 26 candidate biogenic point sources:

- **Biomass power (11 sites, ~18.9 Mt biogenic CO_2/yr of throughput).** Dominated by **Drax** (~12 Mt/yr biogenic CO_2 — the single largest BECCS opportunity in Europe), plus **Lynemouth**, **MGT Teesside**, **Wilton 10 / Sembcorp Teesside**, **Margam (Port Talbot)**, and a cluster of ~40 MW plants (Snetterton, Sleaford, Brigg, Kent/Sandwich, Ince/Mersey). These anchor **BECCS** in East Anglia and the Humber/Teesside and Mersey clusters.
- **Waste-to-energy (13 sites, ~3.0 Mt biogenic CO_2/yr).** The urban EfW fleet — Runcorn, Cory Riverside/Belvedere, Edmonton, Lakeside, Earls Gate (Scotland) — anchors **WtE + CCS** across the English cities and central belt.
- **Anaerobic digestion (existing UK biogas cluster).** Hundreds of farm and food-waste digesters already operate; retrofitting biogas-upgrading CO_2 vents is among the cheapest capture available and anchors **AD + CCS** across the livestock regions.

Crucially, these anchors sit close to the **CO_2 storage hubs now in construction**: the **Northern Endurance Partnership / East Coast Cluster** (Teesside–Humber) and **Liverpool Bay / HyNet** (Mersey), with **Acorn** (Scotland) and **Viking** (Humber) planned. In the Atlas, **all 41 UK regions have "good" (economically reachable) access to gaseous-CO_2 storage** — a far higher share than most of Europe — because these offshore stores are within short transport reach. Modelled delivered CO_2-transport cost is low (~35–65/*tCO₂*, *median* 50), reflecting short routes to the North Sea. This is what makes the CO_2-capture pathways (Scenario A) robust across the whole country.

3. Total CO_2 removal potential by pathway — two storage scenarios

The decisive question for UK BiCRS is **what the CO_2 can be stored as**. Capturing CO_2 as a gas and piping it to an offshore reservoir (BECCS, WtE+CCS, AD+CCS) uses storage the UK is already building. Sequestering **biomass slurry or bio-oil** instead requires a store that takes liquids/solids — in the UK, that means **salt caverns**, which are not yet developed or permitted for this purpose. The two scenarios below bracket that choice.

Scenario A — gaseous-CO_2 storage only (near-term floor: ~8.1 Mt CO_2/yr)

Only CO_2-capture pathways (routing to the offshore hubs) plus storage-independent biochar.

Pathway	Regions	CDR potential	CDR efficiency	Indicative cost
WtE + CCS (capture on energy-from-waste)	18	**4.1 Mt CO_2/yr**	~55%	~$100–200/t
AD + CCS (biogas upgrading capture)	9	**2.3 Mt CO_2/yr**	~37%	$145–300/t
BECCS (biomass power + capture)	2	**1.6 Mt CO_2/yr**	~80%	$200–225/t
Biochar (storage-independent)	1	**0.1 Mt CO_2/yr**	~high	~$100–200/t
No viable pathway	11	—	—	—
Total	41	**~8.1 Mt CO_2/yr**		

The **11 "no viable pathway" regions are the wet livestock areas without an existing digester** — Northern Ireland, Devon, Dorset & Somerset, Cornwall, Cumbria, Teesside, and most of Scotland. Wet manure cannot be combusted or buried, and with no gaseous-CO_2-only route for it, it has *no* removal option here unless anaerobic digestion is built (moving it into AD+CCS) or injection storage becomes available (Scenario B).

Scenario B — with salt-cavern injection storage (~14.9 Mt CO_2/yr)

Developing salt caverns for biomass-slurry / bio-oil injection adds two pathways and rescues the livestock regions. Injection becomes the largest single pathway.

Pathway	Regions	CDR potential	CDR efficiency	Indicative cost
Biomass waste injection (slurry → salt cavern)	8	**5.6 Mt CO_2/yr**	~90%	$125–285/t
WtE + CCS	18	**4.1 Mt CO_2/yr**	~55%	~$100–200/t
AD + CCS	9	**2.3 Mt CO_2/yr**	~37%	$145–300/t
BECCS	2	**1.6 Mt CO_2/yr**	~80%	$200–225/t
HTL bio-oil (bio-crude → salt cavern)	4	**1.3 Mt CO_2/yr**	~50%	$200–400/t
Total	41	**~14.9 Mt CO_2/yr**		

Salt caverns unlock ~6.8 Mt CO_2/yr (+84%), almost entirely in the wet livestock regions. But this storage is **prospective** — the UK has substantial salt-cavern resource (Cheshire, Teesside, East Yorkshire, Lancashire, Dorset, and Islandmagee in Northern Ireland) and bio-oil/biomass storage in salt caverns is a credited method, yet **no UK site is developed or permitted for it today**. Treat Scenario B as developable potential contingent on that storage being established.

How to read this. Two points for prioritisation:

- **Biggest and lowest-regret, in both scenarios: WtE + CCS.** ~4.1 Mt/yr across 18 regions — MSW is the UK's largest biogenic waste stream, concentrated in cities that already run energy-from-waste plants. It removes carbon *and* decarbonises waste the UK must manage anyway, and retrofits an existing urban fleet on storage that already exists. **BECCS** is the highest-efficiency capture route (~80%); **Drax alone** represents removal of the same order as either scenario's total if fully captured — the single most consequential UK decision.
- **The livestock question is a storage question.** Whether the UK's abundant wet manure (the west and north) can do BiCRS at scale hinges on either building anaerobic digestion (AD+CCS, gaseous route) or developing salt-cavern injection. Absent both, those ~11 regions have no near-term pathway.

Bottom line for the UK

The UK's BiCRS advantage is **an existing combustion/AD fleet to retrofit plus the best offshore CO_2 storage access in Europe** — which makes the ~8 Mt/yr of CO_2-capture removal (WtE+CCS, BECCS, AD+CCS) robust and near-term. The **additional ~7 Mt/yr sits in the wet livestock regions and hinges on a storage decision**: developing salt caverns for biomass/bio-oil injection (or building out anaerobic digestion) is what converts that latent potential into deliverable removal. The fastest, highest- quality tonnes come from capturing CO_2 on the plants already clustered around the Teesside/Humber, Mersey, and central-belt hubs; the strategic upside comes from unlocking injection storage. Explore any region's rationale, ranked alternatives, and delivered costs in the interactive Atlas.

Source: UK BiCRS Atlas (https://hausfath.github.io/bicrs-map-uk/), the UK subset of Frontier's BiCRS Atlas. Manure and crop residues distributed by the UK June Agricultural Census (Eurostat Farm Structure Survey); MSW and biosolids by population; forestry from JRC ENSPRESO — all scaled to UK country totals. Gaseous-CO_2 storage from JRC CO2StoP and curated UK CCS projects; salt-cavern injection sites (Scenario B) are curated and treated as prospective (developable, not yet permitted for bio-oil/biomass storage). Facilities curated with capacity-estimated biogenic CO_2. All figures are screening-level estimates carrying material uncertainty and are not a substitute for project-level diligence. Grounded in Frontier's BiCRS purchasing point of view.